# Uncertainty-Aware Dynamic Multiprototype Representation for Out-of-Distribution Detection in SAR Imagery

Xiaoyan Zhou, Tao Tang, Janne Heikkilä, *Senior Member, IEEE*, Li Liu, *Senior Member, IEEE*, Yuli Sun, *Member, IEEE*, and Gangyao Kuang

***Abstract*—Out-of-distribution (OOD) detection is essential for reliable automatic target recognition (ATR) in open-world scenarios. Although widely studied in optical imagery, research on OOD detection in synthetic aperture radar (SAR) images remains limited. SAR imagery poses unique challenges due to high intraclass variability and low interclass separability, which existing distance-based methods with simplistic class representations fail to handle. To address these issues, we propose two dynamic multiprototype representation methods: dynamic multiprototype clustering (DMPC) and uncertainty-aware dynamic multiprototype learning (DMPL). DMPC is a post hoc method that applies hierarchical clustering to assign adaptive prototypes per class, enabling accurate distance-based OOD scoring without retraining. In contrast, DMPL is an end-to-end framework that enhances discrimination in SAR imagery by improving intraclass compactness and interclass separability. It further incorporates an uncertainty-aware filter that updates prototypes only with low intraclass uncertainty samples, ensuring stable and diverse representations. Extensive experiments on the evaluated SAR-OOD benchmark show that the proposed methods achieve strong average OOD detection performance compared with representative post hoc and training-based baselines, demonstrating the effectiveness of dynamic multiprototype representation for SAR OOD detection. Our code is publicly available at https://github.com/Xiaoyan-Zhou/DMPC-DMPL**

***Index Terms*—Out-of-distribution (OOD) detection, prototype learning, synthetic aperture radar (SAR), uncertainty estimation.**

## I. Introduction

SYNTHETIC aperture radar (SAR) is advantageous due to its ability to observe in all-weather and all-time conditions, which makes it widely applicable in agriculture, industry, and other fields. With the improvement of SAR imaging resolution, automatic target recognition (ATR) has become a key technology [1] in applications such as airport management [2], [3] and maritime surveillance [4], [5]. Most deep learning-based SAR ATR methods are developed under the assumption that training and test data are independent and identically distributed (i.i.d.). However, in practical open-world scenarios, unknown target classes not present in the training data may emerge, or the test data distribution may shift from that of the training set. Such violations of the i.i.d. assumption often lead to unreliable predictions and degraded system performance. Consequently, out-of-distribution (OOD) detection [6], [7] is essential to improve the robustness of SAR ATR as it ensures accurate recognition of in-distribution (ID) classes while rejecting OOD inputs, thereby enabling reliable performance in open-world environments.

OOD detection is formulated as a binary classification problem, different from OOD object detection, which focuses on localizing unknown objects. Existing OOD detection methods can be broadly categorized into post hoc and training-based. Post hoc methods design OOD scoring functions to distinguish ID from OOD samples without retraining the model. Representative examples include maximum softmax probability (MSP) [8], Mahalanobis distance [9], and KNN-based scores [10]. These methods are known for their ease of use, as they do not require modifications to the model. However, their performance is fundamentally constrained by the representation capability of the pretrained model. To address this limitation, training-based methods for OOD detection have been proposed by modifying the loss function [11], [12] and network structure [13]. For example, KNN+ [10] employs supervised contrastive (SupCon) loss [14] to improve feature discriminability, achieving better performance than vanilla KNN. Both KNN and KNN+ belong to the broader family of prototype-based or distance-based methods, in which OOD scores are computed from feature distances to training data prototypes. Subsequently, CIDER [11] and PALM [15] further extend this paradigm by introducing a prototype-based training objective that jointly enforces feature dispersion across classes and compactness within associated prototypes, thereby improving the separation between ID and OOD features in OOD detection.

Building on advances in optical image OOD detection, Inkawhich et al. [16] applied MSP and Mahalanobis score functions to enhance SAR OOD detection. Charles Hill uses Gaussian mixture models combined with MSP and other OOD score functions to realize SAR OOD detection [17]. While

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Xiaoyan Zhou is with the College of Electronic Science, National University of Defense Technology, Changsha 410073, China, and also with the Center for Machine Vision and Signal Analysis, University of Oulu, FI-90014 Oulu, Finland.

Tao Tang, Li Liu, Yuli Sun, and Gangyao Kuang are with the College of Electronic Science, National University of Defense Technology, Changsha 410073, China (e-mail: liuli_nudt@nudt.edu.cn).

Janne Heikkilä is with the Center for Machine Vision and Signal Analysis, University of Oulu, FI-90014, Finland.

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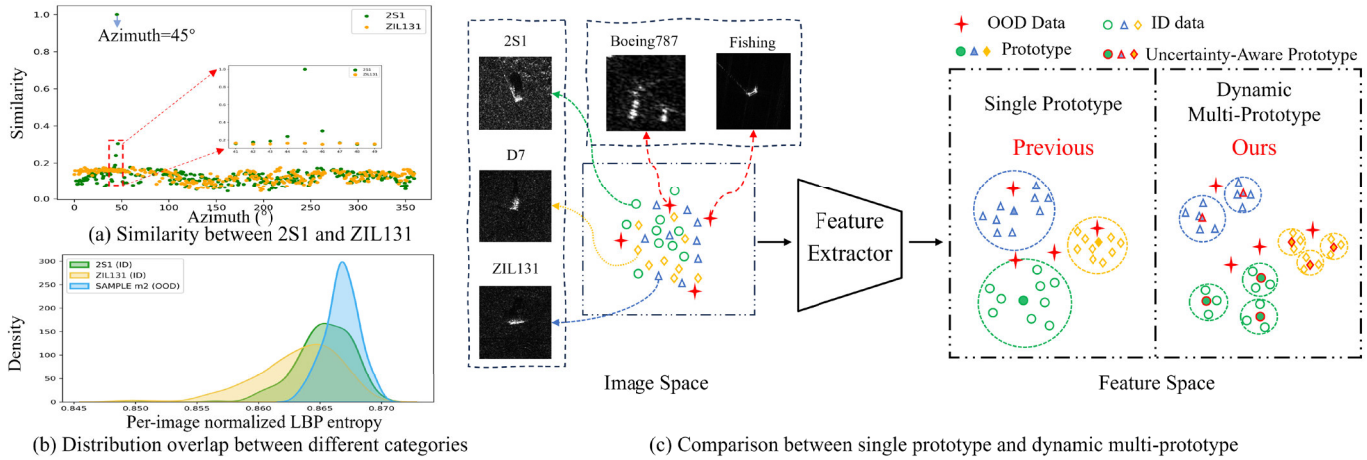



Fig. 1. Motivation of the proposed dynamic multiprototype representation for SAR OOD detection. (a) SAR images exhibit high intraclass variability and low interclass separability due to observation-dependent scattering characteristics. (b) Overlap of the normalized LBP entropy distributions further suggests low interclass separability. (c) Proposed uncertainty-aware dynamic multiprototype representation captures intraclass variability and improves interclass separability, enhancing OOD detection, especially near decision boundaries.

these techniques have been successful for optical imagery, their direct adoption in SAR OOD detection [16], [17], [18] overlooks the fundamental differences in imaging principles between SAR and optical sensors. Optical sensors capture reflected visible light, influenced mainly by surface texture, color, and illumination. In contrast, SAR captures microwave backscattering signatures affected by the target's geometric structure, dielectric properties, and radar imaging geometry [19], [20]. Such coherent scattering mechanisms often introduce scattering ambiguity and clutter interference, which have been widely studied in literature [21], [22], [23] and are known to aggravate feature instability and distribution overlap. These properties lead to two practical challenges for SAR OOD detection on the SAR-OOD benchmark.

1) *Low interclass separability:* Due to SAR imaging effects (e.g., similar scattering structures and geometry-dependent backscatter), leading to partial overlap such as different ID categories or ID and OOD categories [18], [24].
2) *High intraclass variability:* Changes in the observing angle or sensor parameters can lead to significant feature variations within the same target class. This variability is a manifestation of aleatoric uncertainty, which reflects the inherent randomness and noise in the data due to varying imaging conditions.

These phenomena have been well captured in benchmark datasets such as moving and stationary target acquisition and recognition (MSTAR) [25], where Kaplan [26] observed that the similarity between SAR images of the same target drops sharply when the azimuth angle changes by more than 5° due to unstable and nonisotropic scattering as shown in Fig. 1(a). In addition to such intraclass variability, SAR imagery also exhibits low interclass separability, as illustrated in Fig. 1(b), where the LBP entropy distributions are plotted to characterize texture statistics and reveal distribution overlap. These characteristics pose significant challenges for OOD detection. Methods that represent each class with a single prototype, such as KNN, KNN+, and CIDER, are prone to under-representing this diversity. PALM alleviates this issue by introducing multiple prototypes, but it assigns a fixed number of prototypes per class and thus remains limited in complex SAR scenarios. Moreover, existing prototype-based approaches overlook intraclass uncertainty, which often leads to prototype collapse and unreliable representations, thereby degrading OOD detection performance.

These limitations motivate a transition from single or fixed multiprototype representations to more adaptive solutions in SAR imagery. To this end, we propose a dynamic multiprototype representation for SAR OOD detection. It adaptively determines the number of prototypes per class according to SAR data characteristics and incorporates uncertainty into prototype learning, enabling robust separation of ID and OOD samples. Fig. 1(c) illustrates the difference between a single prototype and uncertainty-aware dynamic multiprototypes.

Based on this idea, we propose two schemes that incorporate post hoc and training-based methods.

1) *Dynamic Multiprototype Clustering:* A post hoc method that utilizes hierarchical clustering to learn prototypes in the feature space of SAR imagery. DMPC is a new OOD score function that calculates the distance between the test sample and the nearest prototype rather than all the training samples.
2) *Uncertainty-Aware Dynamic Multiprototype Learning:* A training-based approach that consists of two key components. The ID feature enhancement module (IFEM) is designed to jointly maximize interclass separability and minimize intraclass variance, thereby improving the quality of feature representations for distance-based SAR OOD detection. The prototype optimization module (POM) performs uncertainty-aware refinement of prototypes by filtering out samples with high intraclass uncertainty based on entropy over distance-based probability. This selective update strategy mitigates prototype collapse, enhances update stability, and maintains intraclass diversity during training. In this article, we systematically study the SAR OOD detection

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task from the perspective of post hoc and training. The key contributions of our work can be summarized as follows.

1) We propose dynamic multiprototype representation tailored to low interclass separability and high intraclass variability in SAR imagery.
2) We develop DMPC, a clustering-based post hoc OOD detection method that requires less storage, lower computation, and provides more stable representations than the KNN score function.
3) We design DMPL, an end-to-end training framework that incorporates uncertainty-aware prototype learning to enhance feature discriminability and prototype reliability.
4) Our method achieves the best average performance among the compared post hoc and training-based methods, providing a new perspective for SAR OOD detection.

The remainder of this article is organized as follows. Section I reviews related work. Section III presents the methodology. Section IV reports experimental results, analysis, and discussion. Finally, Section V provides the conclusion.

## II. Related Work

### A. OOD Detection

OOD detection aims to identify ID samples and reject OOD samples [27], which is essential in safety-critical applications. The key challenges in OOD detection are: 1) designing a score function to separate data during the testing phase, where methods inspired by this are known as post hoc approaches and 2) representing data effectively during the training phase, referred to as training methods.

Regarding the post hoc method, Hendrycks and Gimpel introduced the MSP [8] to identify OOD samples, where the maximum probability of ID samples is higher than that of OOD samples. Subsequently, Liang et al. [28] proposed ODIN, which employs temperature scaling and input perturbation to enhance the reliability of probability estimates. Hendrycks et al. [29] further showed that the maximum logit (MaxLogit) score function outperforms MSP in large-scale OOD detection tasks. Building on this, Zhang and Xiang [30] analyzed the gap between MSP and MaxLogit and then proposed DML, which decouples MaxLogit into MaxNorm (the maximum feature norm) and MaxCosine (the maximum cosine similarity between the features and classifier). In parallel, methods based on feature distances, such as KNN [10] and Mahalanobis [31] distance, have been proposed for OOD detection. KNN is notable for the use of nonparametric techniques to measure feature similarity. In addition, some works have explored training strategies for OOD detection, such as incorporating outlier exposure during training [32] or leveraging advances in representation learning [13]. For example, Tack et al. [33] introduced contrastive learning into OOD detection by constructing image pairs using samples and their distributionally shifted augmentations. Huang et al. [13] propose soft cluster-aware equivariant contrastive learning to learn more discriminative semantic features. To explore OOD detection performance in the scenario where only unlabeled ID data is available during training, Sehwag et al. [12] proposed SSD, which employs supervised contrastive learning loss (SupCon) [14] and Mahalanobis [31] distance to enhance ID/OOD separability. Similarly, Sun et al. [10] proposed KNN+, which leverages SupCon and a KNN score function. KNN+ outperforms SSD due to its nonparametric score function, offering better performance in OOD detection. However, methods relying solely on contrastive loss functions remain suboptimal for OOD detection, as they struggle with weak separation between ID and OOD data. To address this issue, Ming et al. [11] proposed CIDER, which utilizes the von Mises–Fisher (vMF) distribution to model unit-norm features and introduces a novel loss function to optimize hyperspherical embeddings into a mixture of vMF distributions.

With the development of OOD detection methods in optical image processing, some of the aforementioned methods have been applied under different scenarios in SAR. For example, Inkawhich et al. [6] used large, diverse, and unlabeled auxiliary data as OOD data and SAMPLE [34] as ID data to train an accurate model, and then utilized ODIN [28] for SAR OOD detection. To improve OOD detection abilities during deployment, Inkawhich et al. [7] proposed a learn-online workflow, which combines offline training and online tuning with OOD data collected from the environment. Inspired by LogitNorm [35], Zhou et al. [18] use evidential deep learning (EDL) to mitigate the overconfidence problem in SAR OOD detection. Gao et al. [24] propose a unified framework for OOD detection and open set recognition (FODOR), where they utilize the same SAR OOD data as in [18]. However, the SAR OOD detection methods above adopt most methods in optical image processing, ignoring the unique characteristics of SAR images.

### B. Prototype Learning

Prototype learning aims to use a set of samples or features to represent the whole dataset. The KNN algorithm is one of the earliest methods in prototype learning [36]. Due to the heavy storage burden and high computational requirements of KNN, learning vector quantization (LVQ) methods were proposed to address this problem. There are two main categories of LVQ: 1) designing prototype updating rules during training [37], [38], [39] and 2) learning prototypes through parameter optimization by designing loss functions [40], [41], [42]. However, these methods first extract features by manually designing feature extractors such as LBP [43] or SIFT [44], and then perform prototype learning; the features are limited by human domain knowledge, potentially failing to capture all the information in the image. Consequently, these prototype learning methods are often inferior to convolutional neural networks in image classification tasks.

Recently, many works have integrated feature representation and prototype learning into end-to-end neural networks [15], [45], [46]. For example, Wen et al. [45] proposed center loss to help CNN models learn discriminative features, where the center can be viewed as a class prototype. Subsequently, prototype learning has been employed to address few-shot learning [47], incremental learning [48], and OOD detection [11], [15]. For OOD detection using prototype learning, CIDER [11]

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utilizes a single prototype to model each class in the data, while PALM [15] employs multiple prototypes. PALM is the work most closely related to our method in optical image OOD detection. However, the number of prototypes per class in PALM is fixed, whereas we demonstrate that a dynamic number of prototypes is more suitable for SAR OOD detection in our experiments. Furthermore, PALM is a training method combined with other OOD score functions, such as KNN, to realize OOD detection, while we propose an OOD score function and training methods based on the DMPL concept.

### C. Uncertainty Estimation

Uncertainty is crucial for assessing the reliability of results, particularly in the realm of trustworthy artificial intelligence. Uncertainty estimation is also used for OOD detection tasks [49], [50], [51] because the OOD data have higher uncertainty than ID data, and the uncertainty estimation method can mitigate the overconfident prediction of deep neural networks [18]. Uncertainty is typically quantified through metrics such as entropy and variance. For instance, Monte Carlo dropout (MC-dropout) [51] implements dropout during the testing phase to generate multiple predictions, using the variance of these predictions as an estimate of uncertainty. Conversely, the deep ensemble [50] method trains several models with different initial random weights to capture prediction uncertainty. Both variance and entropy can serve as an uncertainty metric in MC-dropout and deep ensemble. Shannon entropy is a fundamental measure for estimating uncertainty, characterized by several crucial properties: 1) it is nonnegative and maximal for uniform distributions, indicating maximum uncertainty when outcomes are equally probable; 2) it is continuous, ensuring that small changes in probabilities yield small changes in entropy; 3) it equals zero for certain outcomes, highlighting no uncertainty when an outcome is known; and 4) it is permutation-invariant, meaning it depends solely on the values of probabilities, not on their order. These attributes make Shannon entropy a robust tool across various scientific fields.

Other approaches, such as EDL [52], [53], construct the loss function based on the Dempster–Shafer theory and the Dirichlet distribution, enabling the model to estimate the parameters of the Dirichlet distribution, which is a distribution over categorical probabilities to achieve uncertainty estimation. Compared to methods like MC-Dropout and deep ensemble, EDL provides uncertainty estimates with a single forward computation, offering significantly higher computational efficiency. Most existing uncertainty estimation methods [51], [52] focus on quantifying interclass uncertainty, which pertains to distinguishing between different classes. However, intraclass uncertainty, which means the variability within a single class, has received relatively less attention [54]. Intraclass uncertainty is critical for tasks such as multiprototype learning, where models capture the diversity within classes to improve representational capability, especially when dealing with high intraclass variability or noisy data.

## III. Methodology

The main idea is to utilize the distance between the nearest prototype and a test sample to determine whether the test sample belongs to the ID data or not. Considering the high intraclass variability and low interclass separability of SAR imagery, we propose two dynamic multiprototype representation methods: DMPC and uncertainty-aware DMPL. DMPC is a post hoc method that applies hierarchical clustering to derive an adaptive number of prototypes per class, thereby capturing diverse SAR feature distributions and enabling accurate distance-based OOD scoring. DMPL is a training method that initializes prototypes from DMPC and further optimizes them with a novel loss function in an end-to-end framework, incorporating uncertainty-aware prototype updates for stable and diverse representations.

### A. Preliminaries

This section introduces key concepts related to SAR OOD detection. Consider a $M$-class classification task with $N$ samples training dataset $\mathcal{D}_{\text{train}} = \{(\boldsymbol{x}_i, \boldsymbol{y}_i)\}$ drawn i.i.d. from the joint data distribution $\mathcal{P}_{\mathcal{X}\mathcal{Y}}$, where $\boldsymbol{x}_i \in \mathcal{X}$ represents the input data in the input space $\mathcal{X}$, and $\boldsymbol{y}_i \in \mathcal{Y}$ denotes the corresponding labels in the label space $\mathcal{Y}$. An OOD sample is defined as one drawn from a distribution $\mathcal{P}_{\mathcal{X}'\mathcal{Y}'}$ that differs from $\mathcal{P}_{\mathcal{X}\mathcal{Y}}$, that is, $\mathcal{P}_{\mathcal{X}\mathcal{Y}} \neq \mathcal{P}_{\mathcal{X}'\mathcal{Y}'}$. Thus, OOD detection can be categorized as follows.

1) *Semantic shift*: Where the label distribution $\mathcal{P}_{\mathcal{Y}}$ changes while the conditional distribution $\mathcal{P}_{\mathcal{X}|\mathcal{Y}}$ remains fixed.
2) *Covariate shift*: Where the conditional distribution $\mathcal{P}_{\mathcal{X}|\mathcal{Y}}$ changes while the label distribution $\mathcal{P}_{\mathcal{Y}}$ remains fixed.

In this work, we focus on the semantic shift setting for SAR OOD detection. Furthermore, in our SAR OOD detection task, only ID data is accessible during training. During testing, OOD detection is formulated as a binary classification task, where level set estimation is used to decide whether an input sample $\boldsymbol{x}_i$ belongs to ID or OOD

$$G_{th}(\boldsymbol{x}_i) = \begin{cases} \text{ID}, & \text{if } S(\boldsymbol{x}_i) \geq th \\ \text{OOD}, & \text{if } S(\boldsymbol{x}_i) < th \end{cases} \tag{1}$$

where $th$ denotes the threshold, and $S(\boldsymbol{x}_i)$ is a score function.

### B. Post Hoc Method: DMPC-Based OOD Score Function

DMPC is a distance-based score function that is model-agnostic, as summarized in Algorithm 1. Unlike KNN [10], which calculates the distance between all training samples and the test sample, DMPC computes the distance between prototypes and the test sample. Therefore, DMPC comprises two stages: clustering-based prototype generation and OOD detection. For clustering, the single linkage measures the distance between the closest elements in clusters, formulated as

$$D(C_1, C_2) = \min_{\boldsymbol{x}_1 \in C_1,\ \boldsymbol{x}_2 \in C_2} D(\boldsymbol{x}_1, \boldsymbol{x}_2) \tag{2}$$

where $C_1$ and $C_2$ denote clusters. The Euclidean distance $D(\boldsymbol{x}_1, \boldsymbol{x}_2)$ can be formulated as follows:

$$D(\boldsymbol{x}_1, \boldsymbol{x}_2) = \|\boldsymbol{x}_1 - \boldsymbol{x}_2\|_2. \tag{3}$$

Equation (2) merges clusters according to the nearest pairwise sample distance, which allows hierarchical clustering to capture chain-like or nonglobular structures in SAR feature space.

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**Algorithm 1** OOD Detection With DMPC

**Clustering-based Prototype Generation Stage:**
**Require:** Training dataset $\mathcal{D}_{\text{train}} = \{(\boldsymbol{x}_i, y_i)\}_{i=1}^{N}$ with $M$ classes, pretrained feature extractor $\phi$, total number of prototype $K$
**for** each sample $\boldsymbol{x}_i$ in the training data $\mathcal{D}_{\text{train}}$ **do**
  collect feature-label array $\mathcal{Z}_N = \{(\boldsymbol{z}_1, y_1), (\boldsymbol{z}_2, y_2), \ldots, (\boldsymbol{z}_N, y_N)\}$, where $\boldsymbol{z}_i = \phi(\boldsymbol{x}_i)$
**end for**
Compute the distance matrix $D$ for $\mathcal{Z}_N$ based on Equation (3)
Apply single linkage hierarchical clustering on $D$ to obtain a dendrogram $T$ based on Equation (2)
Cut the dendrogram $T$ to form $K$ clusters $\{C_1, C_2, \ldots, C_K\}$
**for** each cluster $C_k$ with $n_k$ feature vectors **do**
  Compute the cluster center $\boldsymbol{\mu}_k = \frac{1}{n_k} \sum_{(\boldsymbol{z}_i, y_i) \in C_k} \boldsymbol{z}_i$,
  Compute the prototype label $\hat{y}_k = \arg\max_{m \in \{1,\ldots,M\}} \sum_{(\boldsymbol{z}_i, y_i) \in C_k} \mathbf{1}(y_i = m)$
**end for**
**return** $\{(\boldsymbol{\mu}_1, \hat{y}_1), \ldots, (\boldsymbol{\mu}_K, \hat{y}_K)\}$ as prototypes $\mathcal{P}$
**OOD Detection Stage:**
**Require:** learned prototypes $\mathcal{P}$, pretrained feature extractor $\phi$, test sample $\boldsymbol{x}^*$, threshold $th$
calculate feature vector $\boldsymbol{z}^* = \frac{\phi(\boldsymbol{x}^*)}{\|\phi(\boldsymbol{x}^*)\|_2}$ for test sample
Reorder $\mathcal{P}$ according to the increasing value of $\|\frac{\boldsymbol{\mu}_i}{\|\boldsymbol{\mu}_i\|_2} - \boldsymbol{z}^*\|_2$ as $\mathcal{P}' = \{(\boldsymbol{\mu}_{(1)}, \hat{y}_{(1)}), (\boldsymbol{\mu}_{(2)}, \hat{y}_{(2)}), \ldots, (\boldsymbol{\mu}_{(K)}, \hat{y}_{(K)})\}$
**return** OOD detection decision based on Equation (4)

Based on (1), the decision function of DMPC is given by

$$G(\mathbf{z}^*) = \mathbf{1}\{-r(\mathbf{z}^*) \geq th\}$$
$$r(\mathbf{z}^*) = \left\| \frac{\boldsymbol{\mu}_{(1)}}{\|\boldsymbol{\mu}_{(1)}\|_2} - \boldsymbol{z}^* \right\|_2 \tag{4}$$

where $r(\mathbf{z}^*)$ denotes the distance between the test sample and the nearest prototype, $\boldsymbol{z}^*$ is the normalized feature of the test sample, $\boldsymbol{\mu}_{(1)}$ is the prototype in $\mathcal{P}$ closest to the test sample, and $\mathbf{1}\{\cdot\}$ is the indicator function. Note that $r(\mathbf{z}^*)$ is constructed from the feature-to-prototype distance: smaller $r(\mathbf{z}^*)$ indicates higher similarity to the ID prototypes. Therefore, we use $-r(\mathbf{z}^*)$ as the confidence score, where larger values imply higher ID confidence, and apply the threshold $th$ according to (1). Noticeably, except for the advantages of KNN, DMPC still has the following advantages compared to KNN [10].

1) *Reduced storage requirements:* In the clustering-based prototype generation stage, we only store the features and labels of the prototypes for testing.
2) *Lower computational cost:* During the OOD detection stage, we compute distances only between the prototypes and the test sample, enhancing efficiency.
3) *Improved stability:* Prototypes represent mean features of clusters, whereas KNN relies on individual sample features.

Furthermore, in the clustering-based prototype generation stage, each cluster $C_k$ is assigned a prototype label according to the labels of its member samples. Specifically, for the $k$th cluster, we compute the prototype center as the mean of all feature vectors in the cluster and assign its class label by majority voting. The resulting prototype set is $\mathcal{P} = \{(\boldsymbol{\mu}_k, \hat{y}_k)\}_{k=1}^{K}$. For each class $m$, the class-specific prototype set is defined as

$$\mathcal{P}_m = \left\{\boldsymbol{\mu}_k \mid \hat{y}_k = m,\ k = 1, \ldots, K\right\} \tag{5}$$

where the number of prototypes assigned to class $m$ is $K_m = |\mathcal{P}_m|$. Since $K_m$ is not predefined but determined by clustering and prototype-label assignment, the proposed representation is dynamic.

### C. Training Method: Uncertainty-Aware DMPL

Our proposed DMPL is illustrated in Fig. 2. DMPL consists of two main components: 1) the IFEM and 2) the POM. The objective of the IFEM is to learn features that are strongly discriminative between classes and compact within an associated prototype, while the POM aims to obtain accurate prototypes in the SAR feature space by leveraging an uncertainty filter. Overall, our framework achieves these two goals in an end-to-end manner. During testing, we use the OOD detection stage in Algorithm 1 to distinguish test samples.

*1) ID Feature Enhancement Module:* Feature representation is important in OOD detection, especially for distance-based methods. The high internal cohesion of intraclass data can enhance the performance of distance-based OOD detection. Therefore, the loss function of the IFEM should meet two criteria: 1) learn interclass features to achieve accurate classification and 2) ensure that features of images within the same class tightly cluster around the associated prototype. Accordingly, our proposed loss function for $M$ categories and $N$ training samples is as follows:

$$L(\mathcal{P}, \mathcal{X}, \mathcal{Y}) = -\frac{1}{N} \sum_{i=1}^{N} \sum_{j=1}^{M} y_{ij} \log\left(p_{ij}\right) + \frac{\lambda}{N} \sum_{i=1}^{N} \min_{\boldsymbol{\mu}_j \in \mathcal{P}_{y_i}} \left\|\boldsymbol{\phi}(\boldsymbol{x}_i) - \boldsymbol{\mu}_j\right\|_2^2 \tag{6}$$

where $\boldsymbol{y}_i$ is the label of sample $\boldsymbol{x}_i$, $\mathcal{P}$ denotes the prototype set, and $\mathcal{P}_{y_i}$ denotes the set of prototypes whose assigned label is equal to $y_i$ based on (5). The $\phi$ denotes the feature extractor, $\phi(\boldsymbol{x}_i)$ represents the feature of sample $\boldsymbol{x}_i$, and $\boldsymbol{\mu}_j$ is the $j$th prototype in $\mathcal{P}_{y_i}$ closest to $\boldsymbol{x}_i$. The $\lambda$ in (6) is a hyperparameter that determines the importance of the prototype loss. The parameter $\lambda$ in the joint prototype cross-entropy loss (JPEL) balances interclass discrimination and intraclass compactness. A moderate value of $\lambda$ encourages features to remain discriminative while preventing excessive over-constraint toward prototypes. These intuitive roles are further validated by the parameter analysis presented in Section IV.

The second term of (6) is widely applied in prototype learning [46], [55]. The hard minimum operation yields a continuous and piecewise differentiable objective. For most samples, the nearest intraclass prototype is unique, and the gradient is computed with respect to the selected prototype branch, making the objective optimizable using standard stochastic gradient descent (SGD). Nondifferentiability only occurs at prototype-switching boundaries, where a sample is equidistant from two or more intraclass prototypes, resulting

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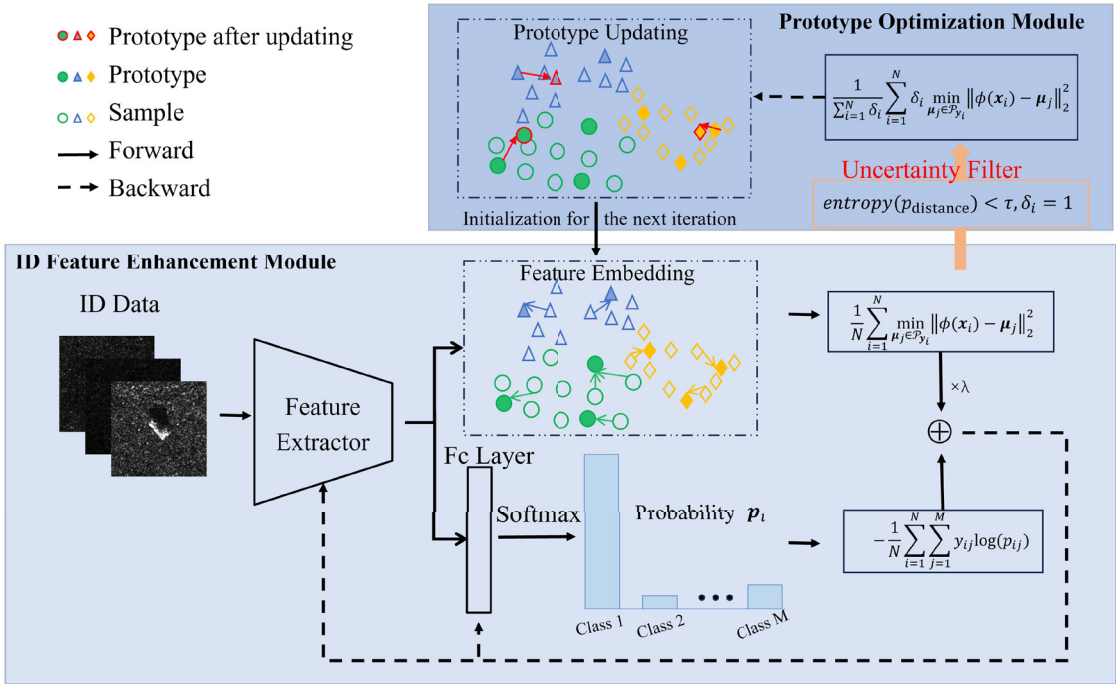



Fig. 2. Overview of our proposed framework for DMPL. We use the prototypes obtained by DMPC. The feature extractor is initialized by the trained classification model. Feature embedding and prototype updating are decoupled by an uncertainty filter: IFEM updates only the model parameters, whereas POM updates only the prototype feature vectors while keeping the prototype labels fixed.

in discontinuities in the gradient. To alleviate the potential instability caused by these boundary samples, we initialize the prototypes using DMPC based on a pretrained feature extractor, which aligns the prototypes with the feature distribution and reduces the likelihood of unstable assignments. Furthermore, the proposed POM employs an uncertainty-aware filter to exclude high intraclass uncertainty samples from prototype updates. Since samples near prototype-switching boundaries typically exhibit high uncertainty, this strategy reduces unstable prototype switching and improves the stability of optimization.

Equation (6), which combines cross-entropy and prototype loss, is named the JPEL. This method of joint supervision can be found in metric learning, such as the joint supervision of softmax loss and center loss for face recognition [45]. The loss in [45] can be viewed as a special case of $L(\mathcal{P}, \mathcal{X}, \mathcal{Y})$ when our prototype set consists of one prototype per class. Furthermore, if the hyperparameter $\lambda = 0$, our loss function degenerates into the cross-entropy loss function commonly used in classification problems.

*2) Prototype Optimization Module:* For prototype updating, both exponential moving average (EMA) and gradient-based methods are usually used. In the gradient-based method, prototypes are directly involved in the gradient descent process, which may result in faster convergence relative to the EMA method, especially if the prototypes need to be adjusted significantly to fit the data distribution. Furthermore, using gradient-based optimization algorithms like SGD to update prototypes ensures consistency in the optimization objective throughout the training process. This means that updates to prototypes are directly linked to the minimization of the loss function, allowing them to adapt more effectively to the training data and task objectives. Therefore, we can use the second term of (6) as the loss function for gradient-based prototype updating, which is illustrated as follows:

$$L = \frac{1}{N} \sum_{i=1}^{N} \min_{\boldsymbol{\mu}_j \in \mathcal{P}_{y_i}} \left\| \boldsymbol{\phi}(\boldsymbol{x}_i) - \boldsymbol{\mu}_j \right\|_2^2. \tag{7}$$

However, (7) simultaneously implements prototype updating and feature embedding, and these two processes are coupled with each other. If multiple intraclass prototypes are very close to or respond to the same sample, they may gradually become more similar, resulting in less discrimination between prototypes. This homogeneity phenomenon reduces the prototypes' ability to cover intraclass variability, limiting the model's capacity to capture and reflect the diversity within the class. We illustrate this situation in Fig. 3.

To address this problem, we propose an entropy-based filtering mechanism that utilizes the distances from samples to intraclass prototypes to gauge the intraclass uncertainty of the SAR sample and only uses low intraclass uncertainty samples for prototype updating. This filtering mechanism effectively mitigates the prototype-switching effect of the min operation on boundary samples, thereby enhancing the smoothness and effective differentiability of the loss function. Furthermore, it can prevent prototypes from frequently shifting due to high intraclass uncertainty samples, thus improving the stability of prototype updates. By updating prototypes using only SAR samples with low intraclass uncertainty, we achieve a good balance of stability and diversity. Specifically, we calculate the distances between the sample and all intraclass prototypes, sort them from smallest to largest, and select the two smallest distances. These distances are converted into probabilities,

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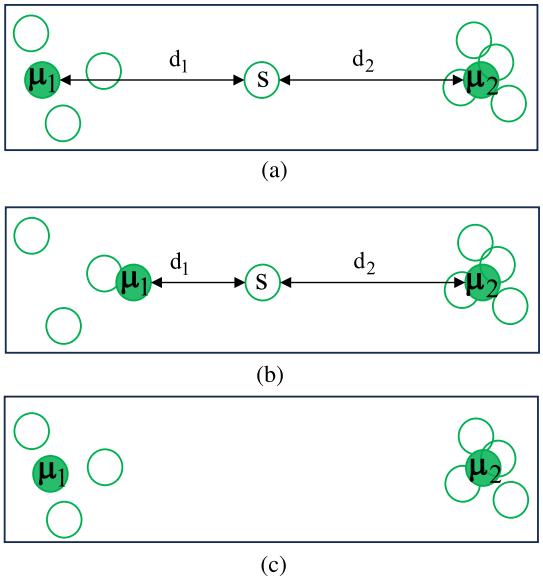



Fig. 3. Influence of prototypes ($\mu_1$ and $\mu_2$) updating with high intraclass uncertainty sample ($S$) in feature space. (a) High intraclass uncertainty samples and intraclass prototypes. (b) Updating prototypes using high intraclass uncertainty samples. (c) Updating prototypes using low intraclass uncertainty samples.

from which we compute Shannon entropy as follows:

$$entropy\left(p_{distance}\right) = -p\log_2 p - (1-p)\log_2(1-p)$$
$$p = \frac{\frac{1}{d_1+\epsilon}}{\frac{1}{d_1+\epsilon}+\frac{1}{d_2+\epsilon}} = \frac{d_2+\epsilon}{d_1+\epsilon+d_2+\epsilon} \tag{8}$$

where the probability $p$ is calculated by an inverse distance. The $\epsilon = 1\times10^{-12}$ is a small positive number used to prevent division by zero and ensure numerical stability. In (8), $d_1$ denotes the Euclidean distance between the sample and the closest intraclass prototype, and $d_2$ denotes the Euclidean distance between the sample and the second closest intraclass prototype. Equation (8) quantifies the intraclass uncertainty, which differs from methods like MC-Dropout that quantify interclass uncertainty [51]. We then build a filter based on the relationship between the threshold and uncertainty value $entropy(p_{distance})$, as follows:

$$\delta = \begin{cases} 1, & entropy\left(p_{distance}\right) < \tau \\ 0, & entropy\left(p_{distance}\right) \geq \tau \end{cases} \tag{9}$$

where $\tau$ denotes the uncertainty threshold and $\delta$ is the filter that determines whether samples are used for prototype updates. The uncertainty threshold $\tau$ controls how strictly high intraclass uncertainty samples are filtered during prototype updates. A smaller $\tau$ results in more aggressive filtering, emphasizing highly reliable samples, whereas a larger $\tau$ allows more samples to contribute to prototype refinement. Accordingly, the loss function in (7), which accounts for the intraclass uncertainty of SAR images in prototype updating, is given by

$$L_p = \frac{1}{\sum_{i=1}^{N}\delta_i}\sum_{i=1}^{N}\delta_i \min_{\mu_j\in\mathcal{P}_{y_i}} \left\|\phi\left(\boldsymbol{x}_i\right)-\boldsymbol{\mu}_j\right\|_2^2. \tag{10}$$

If no sample passes the uncertainty filter (i.e., $\sum_i \delta_i = 0$), the prototype loss $L_p$ is set to zero to avoid division by zero, effectively skipping prototype update for that batch. Based on the loss function in (10) and combined with the SGD optimizer, the prototype parameters are automatically updated. Assuming that the gradient of the loss function with respect to the prototype $\boldsymbol{\mu}$ is $\nabla_{\boldsymbol{\mu}_{k_t}} L_p$, the update formula of prototype $\boldsymbol{\mu}_{k_{t+1}}$ is as follows:

$$\boldsymbol{\mu}_{k_{t+1}} = \boldsymbol{\mu}_{k_t} - \eta\nabla_{\boldsymbol{\mu}_{k_t}} L_p \tag{11}$$

where $\eta$ is a nonnegative number representing the learning rate. A larger $\eta$ leads to faster prototype updating. The uncertainty filter aims to select SAR samples with low intraclass uncertainty for updating, thereby enhancing the stability and diversity of prototypes.

*Training Interaction Between IFEM and POM:* To avoid mutual interference between feature learning and prototype refinement, the IFEM and POM are optimized in a decoupled manner during training. Specifically, the IFEM updates only the feature extractor parameters to enhance interclass separability and intraclass compactness, while keeping prototype parameters fixed. Conversely, the POM updates only prototype parameters using the uncertainty-filtered prototype loss, while freezing the feature extractor. This decoupled optimization strategy stabilizes training and prevents oscillations caused by simultaneous updates of features and prototypes.

## IV. Experiments

In this section, we evaluate the proposed methods on benchmark SAR datasets to validate their effectiveness in OOD detection.

### A. Implementation Details

All experiments are implemented using the PyTorch framework and conducted on a single NVIDIA A100 GPU. To ensure fair comparison, ResNet18 is consistently adopted as the backbone network across all methods. During training, a batch size of 32 is used. For the proposed DMPL method, the network parameters and prototype parameters are optimized using two independent SGD optimizers. Specifically, the backbone network is trained with an initial learning rate of 0.01, momentum of 0.9, and weight decay of $5\times10^{-4}$. The prototype parameters are updated with a separate SGD optimizer using a learning rate of 0.5. All models are trained for 150 epochs to ensure sufficient convergence. For distance-based OOD detection methods, including DMPC, feature vectors are $\ell_2$-normalized before distance computation to stabilize similarity measurement. For KNN and KNN+ baselines, the number of nearest neighbors is set to $K_{nn} = 50$.

### B. Dataset Description and Metrics

We use the SAR-OOD dataset introduced in [18] to evaluate the performance of our method. The SAR-OOD dataset comprises MSTAR SOC [56], SAMPLE (excluding the same categories as in MSTAR SOC) [34], SAR-ACD [57], and FUSAR-ship [58]. MSTAR SOC serves as the ID data, while

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TABLE I

DESCRIPTION OF THE SAR-OOD BENCHMARK. THE ID TRAINING SET USES ALL-AZIMUTH SAR IMAGES CAPTURED AT A 17° DEPRESSION ANGLE, WHILE THE ID TEST SET USES ALL-AZIMUTH IMAGES CAPTURED AT A 15° DEPRESSION ANGLE

| ID / OOD | Task | Datasets | Category | Image |
|---|---|---|---|---|
| ID | Train | MSTAR SOC | 2S1, BMP2, BRDM_2, BTR_60,BTR70, D7, ZIL131, T62, T72, ZSU_23_4 | 2747 |
| ID | Test | MSTAR SOC | 2S1, BMP2, BRDM_2, BTR_60,BTR70, D7, ZIL131, T62, T72, ZSU_23_4 | 2425 |
| OOD | Test | SAMPLE | m1, m2, m35, m60, m548 | 690 |
| OOD | Test | SAR-ACD | A220, A330, A320321, ARJ21, Boeing737, Boeing787 | 3032 |
| OOD | Test | FUSAR-ship | BulkCarrier, CargoShip, ContainerShip, Dredger, Tanker, Fishing, GeneralCargo | 1110 |

the others serve as OOD data. The dataset[1] is detailed in Table I.

We use ID classification accuracy (ID ACC) to measure performance on ID data, which is shown as follows:

$$\mathrm{ACC} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN}} \tag{12}$$

where TP denotes true positives, TN denotes true negatives, FP denotes false positives, and FN denotes false negatives. For OOD data, we employ the area under the receiver operating characteristic curve (AUROC) and the false-positive rate (FPR) as evaluation metrics. A higher AUROC value indicates a stronger ability of the model to distinguish between positive and negative classes. The FPR is formulated as follows:

$$\mathrm{FPR} = \frac{\mathrm{FP}}{\mathrm{TN} + \mathrm{FP}} \tag{13}$$

where the TPR is defined as

$$\mathrm{TPR} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}}. \tag{14}$$

The FPR focuses on instances where negative classes (i.e., OOD samples) are incorrectly classified as positive. In OOD detection, FPR95 is commonly used, representing the FPR at a 95% TPR. A low FPR95 is desirable for SAR OOD detection; for simplicity, we refer to FPR95 as FPR in this article. A good SAR OOD detection method should satisfy: 1) high ID ACC and 2) high AUROC and low FPR. Therefore, we comprehensively verify the algorithm's performance from these two aspects.

In addition to standard OOD detection metrics such as AUROC and FPR, we report the area under the risk-coverage curve (AURC) to evaluate OOD-aware selective prediction performance. AURC measures the tradeoff between prediction risk and coverage, and a lower AURC indicates better reliability under selective prediction. For the OOD detection task, all ID and OOD test samples are first sorted by an ID-likeness score in descending order, where a higher score indicates that the sample is more likely to be ID. At coverage $c = k_{\mathrm{AC}}/n$, the model accepts the top-$k_{\mathrm{AC}}$ ranked samples. Here, $n$ is the total number of evaluated samples and $k_{\mathrm{AC}}$ denotes the number of accepted samples after ranking. The selective risk is then defined as the OOD contamination rate among these accepted samples

$$R(k_{\mathrm{AC}}) = \frac{1}{k_{\mathrm{AC}}} \sum_{i=1}^{k_{\mathrm{AC}}} \mathbf{1}\left[x_{(i)} \text{ is OOD}\right] \tag{15}$$

[1]https://github.com/Xiaoyan-Zhou/SAR-OOD-Detection-Data

TABLE II

OOD-AWARE SELECTIVE PREDICTION PERFORMANCE USING THE AURC METRIC

| Mode | Method | AURC | | | |
|---|---|---|---|---|---|
| | | SAMPLE | SAR-ACD | FUSAR-ship | Average |
| Post-hoc | MaxNorm | 0.2541 | 0.4817 | 0.1955 | 0.3105 |
| | MSP (ICLR 2017) | 0.0361 | 0.2088 | 0.1097 | 0.1182 |
| | ODIN (ICLR 2018) | 0.0373 | 0.2127 | 0.0969 | 0.1156 |
| | Energy (NIPS 2020) | 0.0568 | 0.2188 | 0.1261 | 0.1339 |
| | GradNorm (NIPS 2021) | 0.0924 | 0.3724 | 0.1387 | 0.2012 |
| | MaxLogit (ICML 2022) | 0.0534 | 0.2168 | 0.1259 | 0.1320 |
| | KNN (ICML 2022) | 0.0331 | 0.2052 | 0.0944 | 0.1109 |
| | DML (CVPR 2023) | 0.1505 | 0.3834 | 0.1649 | 0.2329 |
| | DMPC (ours) | 0.0296 | 0.2054 | 0.0880 | **0.1076** |
| Training | KNN+ (ICML 2022) | 0.0318 | 0.1981 | 0.0633 | 0.0978 |
| | CIDER (ICLR 2023) | 0.0340 | 0.1958 | 0.0633 | 0.0977 |
| | DMPL (ours) | 0.0296 | 0.1952 | 0.0634 | **0.0961** |

where $x_{(i)}$ denotes the sample ranked at position $i$ according to the ID-likeness score. We compute AURC by trapezoidal integration over the risk-coverage curve for $c \in [0, 1]$. The random baseline corresponds to the overall OOD ratio in the evaluation set.

### C. Experimental Results

In this section, we compare our methods **DMPC** and **DMPL** with several popular OOD detection methods. For post hoc methods, we include the logit-based **MaxLogit** [29], feature norm-based **MaxNorm**, probability-based **MSP** [8], gradient-based **GradNorm** [59], distance-based **KNN** (with $K_{nn} = 50$) [10], and other methods such as **DML** [30], **Energy** [60] (temperature set to 1 in our experiments), and **ODIN** [28]. For training-based methods, we compare with representative approaches, including **KNN+** (with $K_{nn} = 50$) [10], **CIDER** [11], **FODOR** [24], and **PALM** [15]. We reimplement all training-based methods for SAR OOD detection, except FODOR [24], which follows the same data protocol as ours.

The OOD-aware selective prediction performance is reported in Table II, where a lower AURC indicates a better risk-coverage tradeoff. In the post hoc setting, the proposed **DMPC** achieves the lowest average AURC of 0.1076 among all post hoc methods. Compared with the strongest post hoc baseline KNN, DMPC reduces the average AURC from 0.1109 to 0.1076. Specifically, DMPC obtains the best results on SAMPLE and FUSAR-ship, and achieves a comparable result on SAR-ACD, where its AURC is only slightly higher than that of KNN. These results indicate that DMPC can

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TABLE III
COMPARISON WITH OTHER OOD DETECTION METHODS ON THE SAR-OOD DATASET (%). "–" DENOTES THE ABSENCE OF CORRESPONDING RESULTS IN THE REFERENCED PAPER

| Mode | Method | SAMPLE | | SAR-ACD | | FUSAR-ship | | Average | | ID ACC(%) |
|---|---|---|---|---|---|---|---|---|---|---|
| | | AUROC | FPR | AUROC | FPR | AUROC | FPR | AUROC | FPR | |
| Post-hoc | MaxNorm | 31.48 | 100.00 | 71.37 | 57.29 | 73.69 | 60.09 | 58.85 | 72.46 | 98.02 |
| | MSP (ICLR 2017) [8] | 95.53 | 33.77 | 97.12 | 16.59 | 85.14 | 56.31 | 92.60 | 35.55 | |
| | ODIN (ICLR 2018) [28] | 95.01 | 35.80 | 97.29 | 11.25 | 88.48 | 43.15 | 93.60 | 30.07 | |
| | Energy (NIPS 2020) [60] | 86.22 | 97.39 | 95.66 | 23.28 | 82.01 | 52.97 | 87.96 | 57.88 | |
| | GradNorm (NIPS 2021) [59] | 74.80 | 97.68 | 83.48 | 37.43 | 81.47 | 54.32 | 79.92 | 63.15 | |
| | MaxLogit (ICML 2022) [29] | 87.80 | 95.07 | 96.04 | 22.66 | 82.06 | 54.05 | 88.63 | 57.26 | |
| | KNN (ICML 2022) [10] | 96.87 | 17.25 | 97.93 | 12.57 | 88.92 | 54.05 | 94.57 | 27.96 | |
| | DML (CVPR 2023) [30] | 54.64 | 100.00 | 81.41 | 44.99 | 77.47 | 57.93 | 71.17 | 67.64 | |
| | **DMPC (ours)** | 98.55 | 2.03 | 97.80 | 9.73 | 90.63 | 46.67 | **95.66** | **19.48** | |
| Training | KNN+ (ICML 2022) [10] | 97.70 | 14.35 | 99.40 | 1.88 | 97.66 | 10.63 | 98.25 | 8.95 | 98.14 |
| | CIDER (ICLR 2023) [11] | 96.63 | 21.16 | 99.88 | 0.03 | 99.50 | 0.18 | 98.67 | 7.12 | **99.42** |
| | PALM (ICLR 2024) [15] | 99.08 | 3.91 | 97.50 | 9.30 | 98.94 | 6.22 | 98.50 | 6.48 | 99.30 |
| | FODOR (J-STARS 2025) [24] | 99.30 | 1.98 | 95.20 | 10.20 | 97.00 | 8.00 | 97.17 | 6.73 | – |
| | **DMPL (ours)** | 98.56 | 4.06 | 99.99 | 0.00 | 97.71 | 11.35 | **98.76** | **5.14** | 99.38 |

provide more reliable confidence estimation without requiring additional training.

In the training-based setting, the proposed **DMPL** achieves the best average AURC of 0.0961, outperforming KNN+ and CIDER. DMPL obtains the lowest AURC on SAMPLE and SAR-ACD, while achieving nearly identical performance to the best result on FUSAR-ship. Overall, DMPL shows the most favorable risk-coverage tradeoff across the three datasets, demonstrating that the proposed prototype learning strategy can improve the reliability of OOD-aware selective prediction.

The OOD detection results are shown in Table III. The ID ACC in Table III is calculated on the MSTAR SOC test dataset in Table I. All post hoc methods do not involve retraining the model, resulting in the same ID ACC for them. KNN+ is trained with SupCon loss, which constructs positive pairs using samples within the same class. This emphasis on the cohesiveness of samples within the same class in cases of high intraclass variability can lead to neglecting the diversity among samples of the same class. Consequently, the features learned with SupCon may lack strong generalization capabilities, resulting in suboptimal classification accuracy of KNN+. However, CIDER, PALM, and DMPL not only optimize cross-entropy but also require sample features to be embedded around the closest intraclass prototype, which helps the model learn discriminative features beneficial for classification. Our approach, DMPL, demonstrates strong competitiveness in ID recognition, which illustrates the effectiveness of our proposed framework for ID recognition.

As for SAR OOD detection, DMPC and DMPL achieve the best average results among the compared post hoc and training-based methods, respectively. All post hoc methods use the same backbone to extract features. The performance of MaxLogit, MaxNorm, MSP, DML, Energy, ODIN, and GradNorm shows that SAR OOD detection on the SAR-OOD dataset is a challenging task, and distance-based methods like KNN and DMPC demonstrate their ability to address this issue. DMPC reduces the average FPR by 8.5% compared to KNN, significantly outperforming other post hoc algorithms. We use dynamic multiprototype to capture intraclass diversity, while KNN uses the $K$th nearest-neighbor training sample as a single prototype. Thus, the distances calculated by DMPC are more distinguishable and robust than those from KNN, resulting in better OOD detection performance, as shown in Fig. 4(a) and (b). For training-based OOD detection methods, DMPL achieves the best average performance. Comparing DMPL with DMPC, we conclude that the representation learned with JPEL loss is more distinguishable and compact than that obtained from cross-entropy loss, as illustrated in Fig. 4(b) and (f). From the results of DMPL and CIDER, we find that multiprototype is essential for SAR images to explore intraclass diversity. The comparison between DMPL and PALM shows that the dynamic number of each category can better present the characteristics of the data and is beneficial to OOD detection. The results of both post hoc and training-based methods indicate that the dynamic multiprototype strategy plays a key role in SAR OOD detection.

### D. Parameter Analysis

In this section, we analyze the factors that influence DMPC and DMPL. For the DMPC method, we focus on the aspects related to clustering, such as the linkage method, the number of prototypes, and whether to use the fixed or dynamic number of prototypes per class. For the DMPL method, we explore factors related to our framework, for example, the key parameters in loss functions [see (6) and (10)].

*1) Influence of Dynamic Mode, Linkage Method, and Prototype Number on DMPC:* We investigate the effects of DMPC-related factors in detail, and the results are presented in Table IV.

*a) Dynamic versus fixed mode:* The results in Table IV indicate that the dynamic mode is more suitable for DMPC. The dynamic mode can automatically adjust the number of clusters based on the data distribution, offering more flexibility. In contrast, we set the same number of prototypes per class in the fixed mode, for example, three prototypes per class. It is less adaptable to the distribution of samples within each class, potentially leading to issues of over-clustering or under-clustering, resulting in inaccurate prototypes. Therefore, the dynamic mode performs better than the fixed mode, and dynamic clustering is essential in DMPC.

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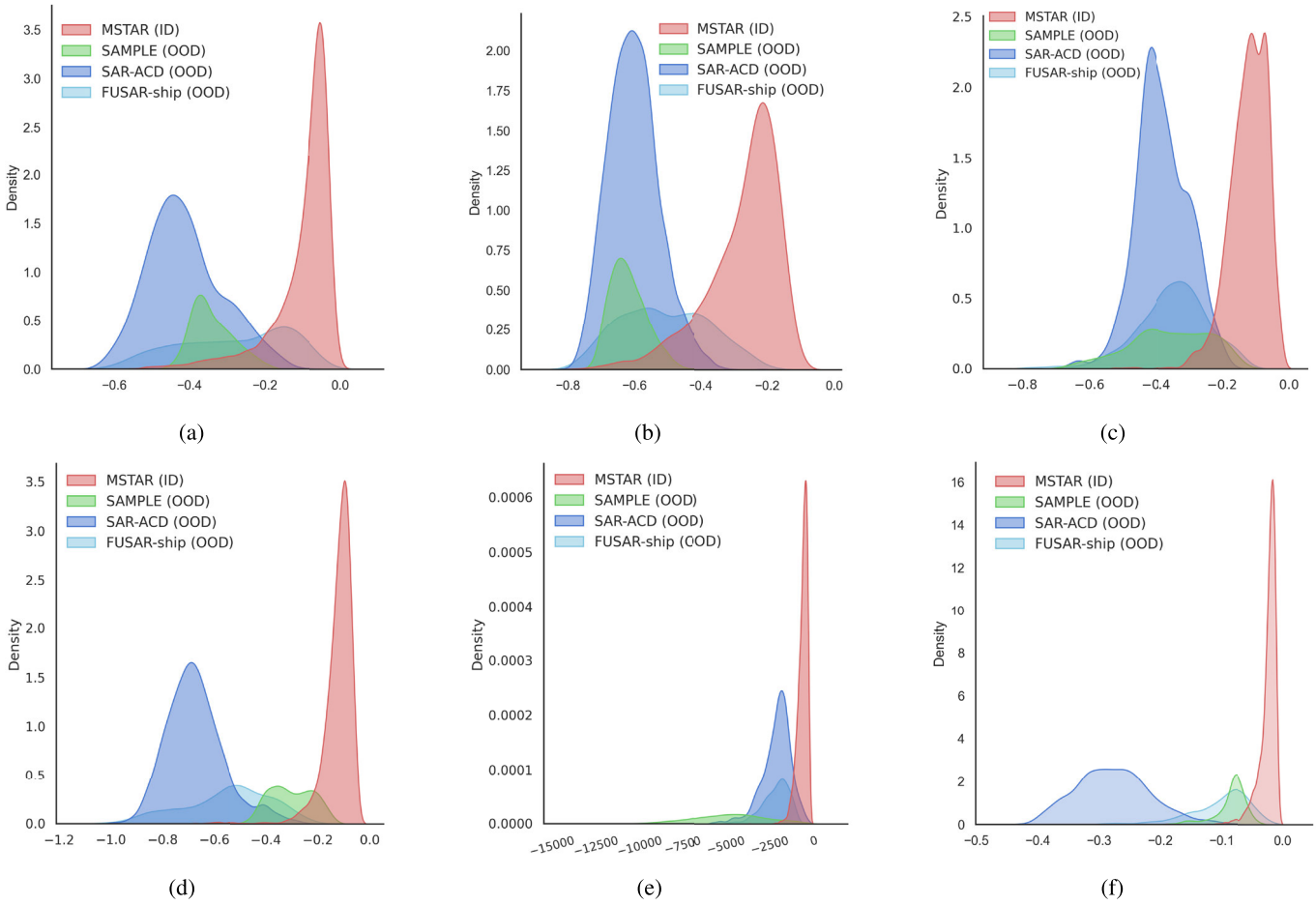



Fig. 4. Distribution of ID and OOD scores calculated by different methods. The $x$-axis of all subplots represents negative distances. Only PALM uses Mahalanobis distance, while others use Euclidean distance. DMPC outperforms KNN, and DMPL is more distinguishable than other training OOD detection methods. (a) KNN. (b) DMPC. (c) KNN+. (d) CIDER. (e) PALM. (f) DMPL.

TABLE IV

INFLUENCE OF LINKAGE METHOD, NUMBER OF PROTOTYPES, AND MODE ON DMPC. $K$ REFERS TO THE TOTAL NUMBER OF PROTOTYPES. VALUES (%) IN THE TABLE REPRESENT THE RESULTS OF DYNAMIC/FIXED PROTOTYPE ASSIGNMENT

| Linkage | K | SAMPLE | | SAR-ACD | | FUSAR-ship | | Average | |
|---|---|---|---|---|---|---|---|---|---|
| | | AUROC | FPR | AUROC | FPR | AUROC | FPR | AUROC | FPR |
| Single | 10 | 97.34 / 97.34 | 11.88 / 11.88 | 97.80 / 97.80 | 12.60 / 12.60 | 88.87 / 88.87 | 53.42 / 53.42 | 94.67 / 94.67 | 25.97 / 25.97 |
| | 20 | 97.60 / 97.48 | 10.29 / 9.86 | 97.83 / 97.75 | 12.30 / 11.84 | 88.99 / 88.90 | 53.06 / 52.61 | 94.81 / 94.71 | 25.22 / 24.77 |
| | 30 | 97.73 / 97.76 | 8.55 / 8.41 | 97.74 / 97.79 | 12.01 / 11.48 | 89.06 / 89.04 | 53.06 / 53.33 | 94.84 / 94.86 | 24.54 / 24.41 |
| | 40 | 98.04 / 97.86 | 6.52 / 6.81 | 97.84 / 97.85 | 10.78 / 10.46 | 89.40 / 89.12 | 52.43 / 54.05 | 95.09 / 94.94 | 23.25 / 23.77 |
| | 50 | 98.18 / 98.09 | 5.36 / 5.94 | 97.58 / 97.66 | 11.81 / 11.81 | 89.64 / 89.70 | 51.35 / 52.25 | 95.13 / 95.15 | 22.84 / 23.33 |
| | 60 | 98.38 / 95.50 | 2.46 / 28.55 | 97.52 / 97.58 | 10.69 / 11.05 | 90.17 / 88.16 | 47.03 / 51.53 | 95.36 / 93.75 | 20.06 / 30.38 |
| | 70 | **98.55** / 95.70 | **2.03** / 27.54 | 97.80 / 97.66 | **9.73** / 10.72 | **90.63** / 88.47 | **46.67** / 51.62 | **95.66** / 93.94 | **19.48** / 29.96 |
| Ward | 10 | 97.34 / 97.34 | 11.88 / 11.88 | 97.80 / 97.80 | 12.60 / 12.60 | 88.87 / 88.87 | 53.42 / 53.42 | 94.67 / 94.67 | 25.97 / 25.97 |
| | 20 | 97.59 / 97.34 | 8.99 / 12.03 | **97.96** / 97.53 | 10.92 / 14.64 | 89.77 / 89.53 | 51.71 / 52.70 | 95.11 / 94.80 | 23.87 / 26.46 |
| | 30 | 97.14 / 95.42 | 11.30 / 32.17 | 97.77 / 97.70 | 12.47 / 13.16 | 90.07 / 88.50 | 50.63 / 54.23 | 94.99 / 93.87 | 24.80 / 33.19 |
| | 40 | 95.76 / 95.79 | 28.26 / 27.97 | 97.79 / 97.80 | 12.96 / 12.73 | 88.91 / 88.91 | 54.32 / 53.96 | 94.15 / 94.17 | 31.85 / 31.56 |
| | 50 | 96.03 / 96.05 | 23.77 / 23.77 | 97.89 / 97.85 | 11.15 / 11.28 | 89.14 / 89.18 | 52.88 / 52.97 | 94.35 / 94.36 | 29.27 / 29.34 |
| | 60 | 96.18 / 96.21 | 23.33 / 23.62 | 97.94 / 97.90 | 11.05 / 11.68 | 89.51 / 89.48 | 52.52 / 53.60 | 94.54 / 94.53 | 28.97 / 29.63 |
| | 70 | 96.41 / 96.34 | 20.58 / 21.88 | 97.94 / 97.83 | 11.28 / 12.24 | 89.83 / 89.68 | 52.61 / 53.24 | 94.73 / 94.61 | 28.16 / 29.12 |

*b) Linkage method:* We conduct experiments with single linkage [see (2)] and ward linkage. The ward linkage in hierarchical clustering is defined as

$$TD_{\{C_1 \cup C_2\}} = \sum_{\boldsymbol{x} \in \{C_1 \cup C_2\}} D\left(\boldsymbol{x}, \boldsymbol{\mu}_{\{C_1 \cup C_2\}}\right)^2 \tag{16}$$

where $TD_{\{C_1 \cup C_2\}}$ denotes the total within-cluster variance after merging clusters $C_1$ and $C_2$, and $\boldsymbol{\mu}_{\{C_1 \cup C_2\}}$ denotes the mean of the merged cluster. The core idea of the ward linkage is to minimize the within-cluster variance during the clustering process by merging clusters that result in the smallest increase in total variance.

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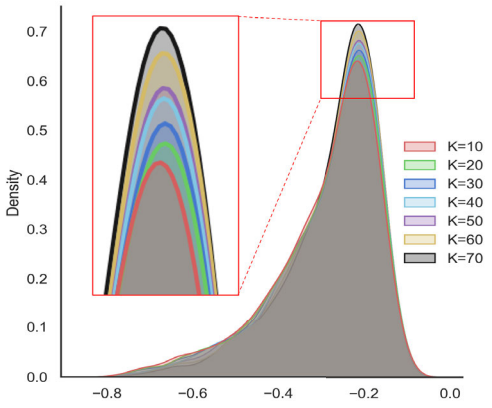



Fig. 5. Distribution of negative distance with different total prototype numbers ($K$).

Our experimental results regarding the linkage methods are illustrated in Table IV. From the results, we conclude that single linkage performs better than ward linkage in the dynamic mode and fixed modes with a small prototype number. SAR images capture the scattering information of targets and are sensitive to observation angles. Specifically, different observation angles can cause variations in the target's contour, shadow, and other features in the image. The similarity between SAR images of the same target decreases as the observation angle differences increase. Furthermore, different targets may exhibit similar SAR image characteristics at specific observation angles, leading to higher similarity between images of different categories at certain angles. This indicates that the distribution of SAR images is not simply globular but rather of nonglobular shapes. Single linkage performs well on nonglobular data. Consequently, compared to ward linkage, single linkage can obtain more accurate cluster centers, thereby achieving superior OOD detection performance.

*c) Prototype number:* From the results of dynamic mode and single linkage in Table IV, we observe that DMPC performance improves as the number of prototypes increases. We illustrate the distribution with different numbers of prototypes in Fig. 5. As the number of prototypes increases, the distribution of negative distances gradually becomes more concentrated within a smaller range, particularly around −0.2. This indicates that the distances from ID samples to their nearest prototypes become more consistent, resulting in a clearer and more compact decision boundary that enhances OOD detection performance. Furthermore, this finding aligns with the motivation of our method in Fig. 1.

Nevertheless, the benefit of increasing the prototype number is not unlimited. After a certain point, the distance distribution tends to stabilize, indicating that additional prototypes provide only marginal improvement in representation capacity. Meanwhile, a larger prototype set increases the computational cost of nearest-prototype distance calculation. Therefore, selecting an appropriate total number of prototypes can strike a balance between performance and computational complexity.

*2) Influence of Key Parameters on DMPL:* We explore the performance of the DMPL model under varying conditions determined by three parameters: the total number of prototypes $K$, the uncertainty threshold $\tau$, and $\lambda$ in the JPEL loss function. Specifically, $K$ varies over {10, 20, 30, 40, 50, 60, 70}; the threshold $\tau$ is set at three levels: 0.85, 0.90, and 0.95; and the hyperparameter $\lambda$ spans a range from 0.1 to 3.0, including the values {0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0, 1.5, 2.0, 2.5, 3.0}. Note that the uncertainty threshold is only effective when the number of prototypes per class is larger than one, as defined in (8). Therefore, for $K = 10$, where only a single prototype is used per class, the SAR OOD detection results remain unchanged regardless of the threshold value, as shown in Fig. 6.

*a) Influence of $\tau$:* The AUROC, FPR, and ID ACC heatmaps presented in Fig. 6 illustrate the optimal results achieved for each combination of $K$ and $\tau$, where the best-performing $\lambda$ value is selected for each scenario. This selection aims to maximize the AUROC and ID ACC while minimizing the FPR, ensuring that the results reflect the optimal configuration under varying $\lambda$ conditions. The main observations are as follows.

1) *AUROC*: We sought the configuration yielding the highest AUROC to ensure optimal discriminative performance between ID and OOD data. The maximum AUROC observed was 99.09 at $K = 60$ and $\tau = 0.9$.
2) *FPR*: A lower FPR indicates a lower rate of false positives. The lowest FPR recorded was 4.55 at $K = 30$ and $\tau = 0.9$.
3) *ID ACC*: We aimed for the highest possible accuracy on ID data. The best ID ACC was 99.51% at $K = 20$ and $\tau = 0.95$.

These findings suggest that an uncertainty threshold of $\tau = 0.9$ consistently provides robust performance across multiple metrics. This threshold, especially when combined with appropriate $K$ values ranging between 20 and 60, optimizes the model's effectiveness in detecting OOD data while efficiently balancing OOD and ID performance. Consequently, we set the uncertainty threshold $\tau = 0.9$ to further explore the influence of $\lambda$ and $K$.

*b) Influence of $\lambda$ and $k$ with $\tau = 0.9$:* Fig. 7 presents the ID ACC heatmap for different values of $\lambda$ and $K$. In (6), $\lambda$ modulates the importance of the regularization term relative to the primary loss function. Specifically, when $\lambda = 0$, the loss function simplifies to the standard cross-entropy loss. As $\lambda$ increases, the model places greater emphasis on intraclass feature cohesion to enhance recognition performance. However, when $\lambda$ exceeds 1.0, recognition accuracy begins to decline. The optimal $\lambda$ achieves a balance between interclass discriminability and intraclass cohesion, thereby improving ID ACC. For instance, when $\lambda = 0.5$ and $K = 40$, the recognition accuracy reaches 99.38%, outperforming the 98.02% accuracy observed when $\lambda = 0$.

The SAR OOD detection results are illustrated in Fig. 8. For most combinations of $\lambda$ and $K$, the DMPL is better than DMPC. This indicates that the DMPL method is beneficial for obtaining better prototypes through the training method. The performance of OOD detection degrades significantly when $K = 10, \lambda = 3.0$. Moderate $\lambda$ values (e.g., 0.1–1.0) typically balance maintaining high AUROC while controlling FPR. Increasing $K$ (i.e., using more prototypes) generally improves

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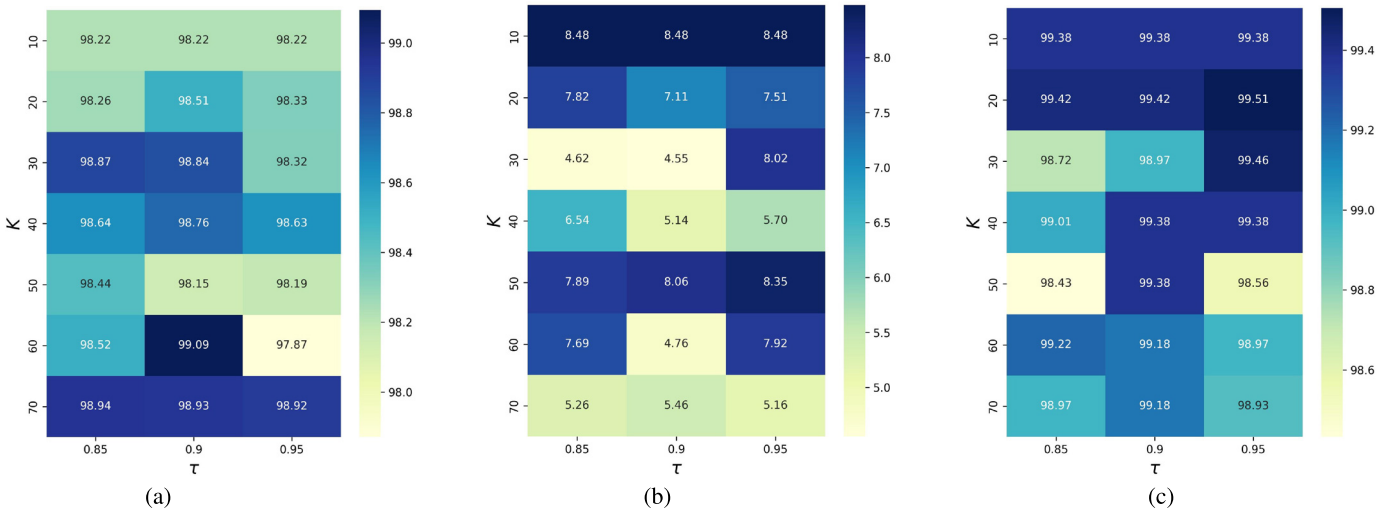



Fig. 6. Average AUROC, Average FPR, and ID ACC of DMPL with different $\tau$, $K$ on the SAR-OOD dataset. The higher AUROC and ID ACC are better, and the lower FPR is better. (a) Average AUROC (%). (b) Average FPR (%). (c) ID ACC (%).

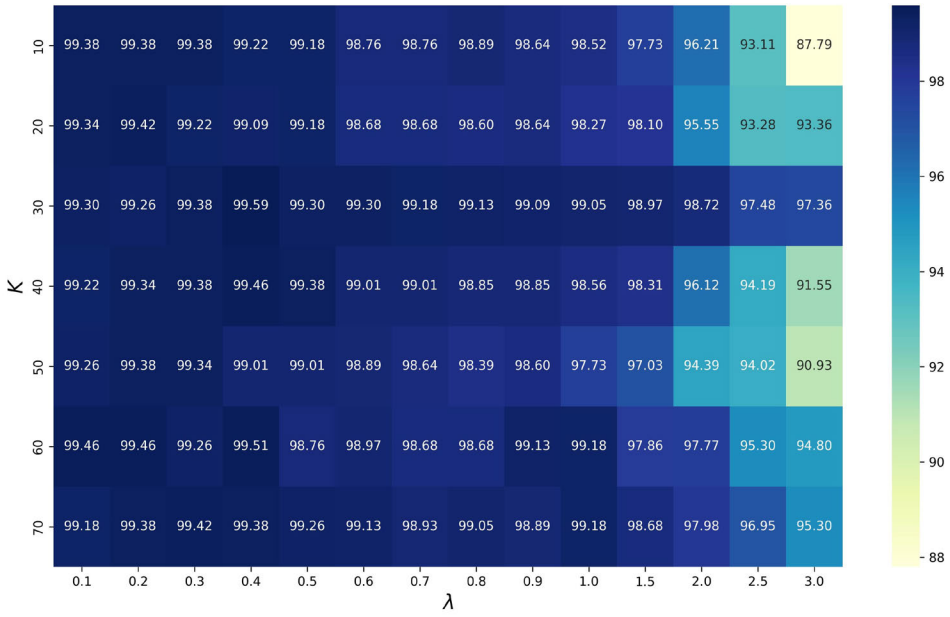


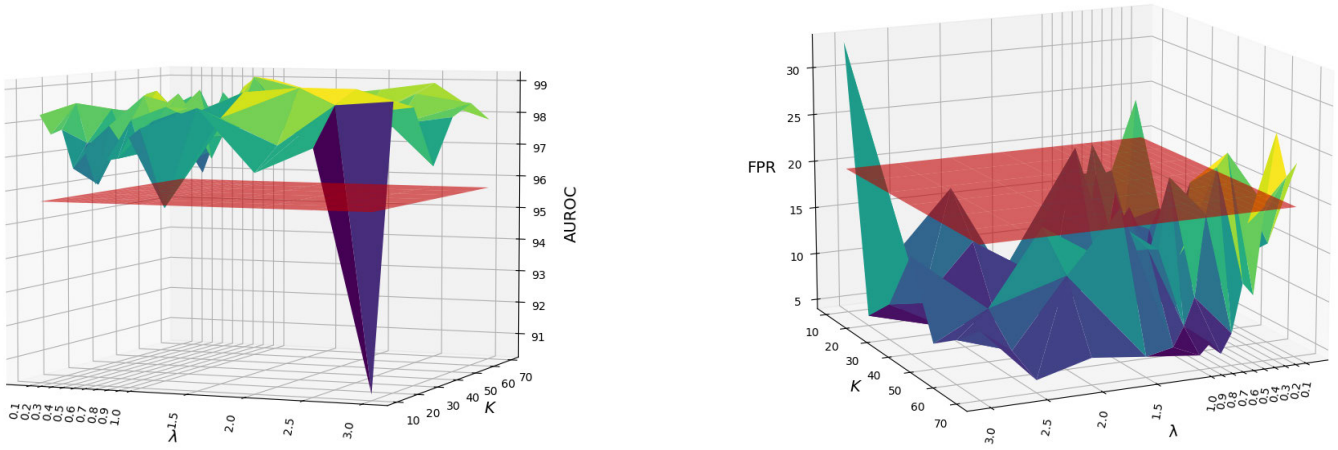

Fig. 7. ID ACC (%) heatmap by $\lambda$ and $K$ while $\tau = 0.9$.

OOD detection performance, particularly with suitable $\lambda$ values. By increasing $K$, the model's interclass discriminability is further optimized, thereby reducing FPR. However, increasing $K$ also increases the storage cost and the computational complexity of nearest-prototype distance calculation. Therefore, $K$ should be selected by balancing OOD detection performance and computational efficiency. For SAR OOD detection on the SAR-OOD dataset, $K = 40$ provides a good empirical tradeoff. Overall, by appropriately adjusting $\lambda$ and $K$, the OOD detection performance can be optimized. Additionally, when $K = 10$ and $\lambda = 3.0$, the AUROC of DMPL is low, the FPR is high, and the ID ACC is low. OOD detection relies heavily on accurate ID representations to differentiate between ID and OOD samples. If the ID ACC is low, the model fails to learn well-defined representations for the ID data. Consequently, the improvement of ID ACC should not be ignored when designing an OOD detection method.

### E. Ablation Study

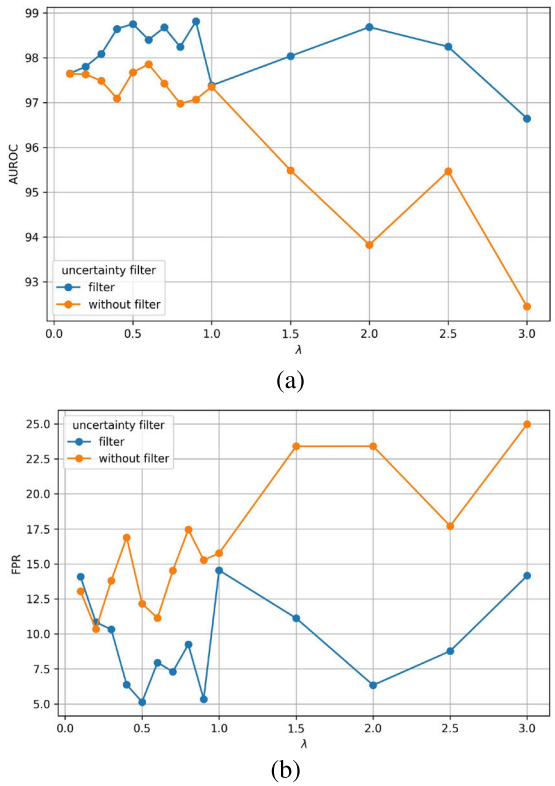
*1) Influence of Uncertainty Filter:* Based on the findings in Section IV-D2, we further study the impact of the uncertainty filter on OOD detection performance when $K = 40$ and $\tau = 0.9$. Fig. 9 presents the comparison of average AUROC and FPR over three datasets, under settings with and without the uncertainty filter. As shown in Fig. 9(a), when $\lambda$ exceeds 1.0, the performance gap becomes more pronounced, indicating the filter's effectiveness in preserving interclass separability even with strong regularization. In Fig. 9(b), the FPR with the uncertainty filter remains significantly lower, which

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(a) (b)

Fig. 8. Average AUROC and Average FPR of DMPL with different $\lambda$ and $K$ values on SAR-OOD dataset. The red plane represents the performance of DMPC in Table III. (a) Average AUROC (%). (b) Average FPR (%).




(a)

(b)

Fig. 9. Performance comparison of OOD detection with and without an uncertainty filter when $K = 40$ and $\tau = 0.9$. (a) Comparison of average AUROC (%). (b) Comparison of average FPR (%).

suggests that without filtering out high intraclass uncertainty SAR samples, prototype updates may be negatively impacted by noisy or ambiguous data, leading to more frequent misclassification of OOD samples as ID. These results demonstrate that integrating an uncertainty filter in prototype optimization not only helps suppress the influence of high intraclass uncertainty samples but also contributes to maintaining the diversity and stability of the learned prototypes. As a result, the model's capability to distinguish SAR OOD from ID samples is enhanced, especially under varying regularization strengths.

*2) Influence of Multiprototype Initialization:* Based on the ablation study in Section IV-E1, DMPL with the uncertainty filter achieves better OOD detection performance under various $\lambda$. To further analyze the influence of dynamic multiprototype initialization under the same uncertainty-aware update setting, we compare three initialization strategies.

1) Dynamic multiprototype initialization obtained from DMPC, where the number of prototypes assigned to each class is determined by the SAR feature distribution.
2) Fixed multiprototype initialization using a normal distribution.
3) Fixed multiprototype initialization using a uniform distribution. All settings use the same DMPL training framework and uncertainty filter. The comparison is conducted with $\lambda = 0.5$, and the results on the SAR-OOD dataset and ID test set are shown in Fig. 10.

As shown in Fig. 10, dynamic multiprototype initialization achieves the best overall tradeoff between OOD detection and ID recognition. Compared with fixed normal initialization, the dynamic initialization substantially improves OOD detection, reducing the average FPR from 48.90% to 5.14%. Fixed uniform initialization achieves competitive OOD detection performance, but its ID accuracy is lower than that of dynamic initialization. These results indicate that data-driven dynamic prototype initialization provides more reliable prototypes for DMPL, leading to stable OOD detection while preserving strong ID recognition performance.

### F. DMPC Versus DMPL: Applications and Insights

The core objective of post hoc-based DMPC and training-based DMPL is to derive accurate prototypes to capture SAR image intraclass diversity, which plays a crucial role in distance-based SAR OOD detection. Notably, DMPL, which refines uncertainty-aware prototypes through model training,

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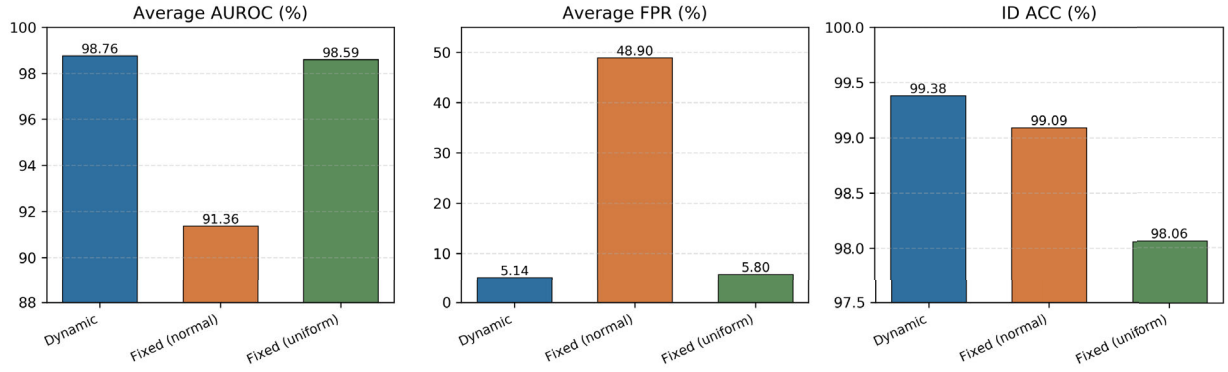



Fig. 10. Dynamic and fixed multiprototype learning in DMPL.

can be viewed as an optimization of the prototypes initially derived by DMPC. As a result, DMPL generally achieves superior OOD detection performance compared to DMPC (see Fig. 8). However, each method serves distinct applications in SAR OOD detection as follows.

DMPC as a post-hoc approach offers distinct advantages for training-free deployment:

1) *DMPC (Post Hoc)*:
   a) *Retrofitting existing SAR ATR systems*: Ideal for scenarios where a pretrained model is already deployed, and there is a need to handle unexpected inputs without retraining.
   b) *Low-cost adaptation*: Well-suited for situations where retraining is infeasible due to computational or time constraints.
   c) *No retraining required*: Allows quick adaptation to unexpected inputs without modifying the entire model, making it computationally efficient and fast to deploy.

DMPL as a training-based approach delivers optimized performance at the cost of additional training resources:

2) *DMPL (Training-Based)*:
   a) *Proactive OOD awareness*: Appropriate for use cases where early detection and awareness of OOD data are essential.
   b) *High-stakes applications*: Effective for scenarios where optimal OOD detection performance is crucial, such as in mission-critical systems.
   c) *Optimized performance*: Particularly advantageous when maximizing OOD detection performance is a priority, and the additional cost of retraining is acceptable.
   d) *Retraining allowed*: Best suited when there are sufficient resources to allow model retraining for enhanced OOD detection accuracy.

While our approach systematically addresses the SAR OOD detection problem and achieves competitive performance among the compared methods, it still requires the manual setting of the total number of prototypes ($K$). Future research could focus on developing methods to automatically learn this critical parameter, further enhancing the adaptability and efficiency of the proposed methods.

## V. Conclusion

To address the lack of consideration of SAR image characteristics in OOD detection, we have introduced a dynamic multiprototype approach, which utilizes the distance between test samples and the nearest prototype to identify OOD data. Building upon this idea, we propose two methods: DMPC and DMPL. DMPC is a post hoc method that utilizes the nonglobular nature of SAR features and dynamically allocates prototypes per class according to data distribution. DMPL is an end-to-end training framework composed of two modules: IFEM and POM. IFEM introduces a JPEL to enhance interclass separability and intraclass compactness, thereby forming feature distributions more suitable for SAR OOD detection. POM refines prototype representations through an entropy-based uncertainty filter, which excludes high intraclass uncertainty samples during updates and mitigates prototype collapse. This strategy preserves prototype diversity and improves update stability. Our methods demonstrate promising results on the SAR-OOD benchmark, underscoring the critical importance of considering the low interclass separability and high intraclass variability of SAR images in OOD detection. This approach enhances the reliability of SAR ATR in an open-world environment and provides a new perspective for applying prototype representation in this challenging domain.

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**Xiaoyan Zhou** received the B.S. degree in electronic information engineering from Chengdu University of Information Technology, Chengdu, China, in 2019, and the M.S. degree in information and communication engineering from the National University of Defense Technology (NUDT), Changsha, China, in 2021. She is currently pursuing the Ph.D. degree with the State Key Laboratory of Complex Electromagnetic Environment Effects, NUDT.

She is also a Visiting Ph.D. Student with the University of Oulu, Finland. Her research interests include remote sensing image processing, uncertainty estimation, and machine learning.



**Tao Tang** received the B.S., M.S., and Ph.D. degrees in information and communication engineering from the National University of Defense Technology, Changsha, China, in 2002, 2006, and 2016, respectively.

He has been an Associate Professor with the School of Electronic Science, National University of Defense Technology. His research interests include synthetic aperture radar target detection and recognition, and remote sensing image registration and feature extraction.



**Janne Heikkilä** (Senior Member, IEEE) received the Doctor of Technology degree in information engineering from the University of Oulu, Finland, in 1998.

Since 2010, he has been a Professor of computer science and engineering with the Faculty of Information Technology and Electrical Engineering, University of Oulu. Since 2024, he has been the Head of the Center for Machine Vision and Signal Analysis (CMVS), Oulu. From 2006 to 2009, he was the President of the Pattern Recognition Society of Finland. Throughout his career, he has been a principal investigator of numerous publicly funded research projects. He has authored or co-authored over 200 peer-reviewed scientific articles. His research interests include 3-D computer vision, machine learning, and computational photography.

Dr. Heikkilä is an IAPR Fellow. He has actively contributed to the academic community as a member of editorial boards and program committees for various international journals and conferences. He is currently an Associate Editor of *IET Computer Vision*.



**Li Liu** (Senior Member, IEEE) received the Ph.D. degree from the National University of Defense Technology (NUDT), Changsha, China, in 2012.

She is currently a Full Professor with NUDT. She has held visiting appointments with the University of Waterloo, Waterloo, ON, Canada, the Chinese University of Hong Kong, Hong Kong, and the University of Oulu, Finland. Her papers currently have over 10 000 citations according to Google Scholar. Her research interests include computer vision, pattern recognition, and machine learning.

Dr. Liu served as a Co-Chair of many International Workshops along with major venues like CVPR and ICCV. She served as the Leading Guest Editor of the special issues for IEEE TRANSACTIONS ON PATTERN ANALYSIS AND MACHINE INTELLIGENCE and *International Journal of Computer Vision*. She also served as an area Chair for several respected international conferences. She currently serves as an Associate Editor for IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS FOR VIDEO TECHNOLOGY and *Pattern Recognition*.



**Yuli Sun** (Member, IEEE) received the Ph.D. degree in information and communication engineering from the National University of Defense Technology (NUDT), Changsha, China, in 2023.

He is currently a Post-Doctoral with the College of Aerospace Science and Engineering, NUDT. He has authored articles in IEEE TRANSACTIONS ON IMAGE PROCESSING, IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS, IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, *ISPRS Journal of Photogrammetry and Remote Sensing*, and *Pattern Recognition*. His research interests cover remote sensing image processing and multitemporal image analysis.



**Gangyao Kuang** received the B.S. and M.S. degrees in geophysics from the Central South University of Technology, Changsha, China, in 1988 and 1991, and the Ph.D. degree in communication and information from the National University of Defense Technology, Changsha, in 1995.

He is currently a Professor with the College of Electronic Science, National University of Defense Technology. His research interests include remote sensing, SAR image processing, change detection, SAR ground moving target indication, and classification with polarimetric SAR images.

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